

Correct call_timestamp and csat_score

The screenshot shows the MySQL Workbench interface with a SQL script in the 'SQL File 1' editor. The script includes comments and SQL commands to update the 'calls' table. The 'calls' table structure is visible in the left sidebar, and the 'Result Grid' at the bottom shows the output of the script.

```
15 response_time CHAR(20),
16 call_duration_minutes INT,
17 call_center CHAR(20)
18 }
19
20 SELECT * FROM calls limit 100;
21
22 SELECT call_timestamp FROM calls LIMIT 10;
23
24 DESCRIBE calls;
25
26 ALTER TABLE calls ADD call_timestamp_new DATE;
27
28 SET sql_safe_updates = 0;
29
30 UPDATE calls SET call_timestamp_new = str_to_date(call_timestamp, '%m/%d/%Y');
31
32 ALTER TABLE calls DROP COLUMN call_timestamp;
33
34 SELECT * FROM calls LIMIT 10;
35
36 ALTER TABLE calls CHANGE call_timestamp_new call_timestamp DATE;
```

The 'Result Grid' shows the output of the script, including the 'calls' table structure and the results of the UPDATE and SELECT statements.

ID	call_name	sentiment	reason	city	state	channel	response_time	call_duration_minutes	call_center	call_timestamp
000	Analise Gardner	Neutral	Billing Question	Detroit	Michigan	Call Center	Within SLA	17	Los Angeles,CA	2020-10-29
000	QSK-72219678-w-102139-KY	Very Positive	Service Outage	Spartanburg	South Carolina	Chatbot	Within SLA	23	Baltimore,MD	2020-10-05
000	017-3802832-A-02015-AD	Negative	Billing Question	Gainesville	Florida	Call Center	Above SLA	46	Los Angeles,CA	2020-10-04
000	126-66807599-63000-6-7	Very Negative	Billing Question	Portland	Oregon	Chatbot	Within SLA	12	Los Angeles,CA	2020-10-17
000	DDU-69481719-O-176402-Fm	Very Positive	Payments	Fort Wayne	Indiana	Call Center	Within SLA	22	Los Angeles,CA	2020-10-17
000	367-9726660-0-22005-66	Neutral	Billing Question	Salt Lake City	Utah	Call Center	Within SLA	25	Baltimore,MD	2020-10-28
000	AZS-65054097-4-85542-PT	Neutral	Billing Question	Tyler	Texas	Chatbot	Within SLA	31	Baltimore,MD	2020-10-16
000	TWV-27007918-5-608789-Bw	Positive	Billing Question	New York City	New York	Chatbot	Below SLA	37	Los Angeles,CA	2020-10-21
000	194-4409118-P-394472-DJ	Very Negative	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore,MD	2020-10-03
000	RLC-64108207-2-28141-VS	Port Ingress	Billing Question	Cincinnati	Ohio	Chatbot	Within SLA	12	Baltimore,MD	2020-10-07

The 'Output' pane shows the results of the script execution, including the 'calls' table structure and the results of the UPDATE and SELECT statements.

```
13 03:55:09 ALTER TABLE calls DROP COLUMN call_timestamp
14 03:55:41 ALTER TABLE calls CHANGE call_timestamp_new call_timestamp DATE
15 03:55:43 DESCRIBE calls
16 03:55:54 SELECT * FROM calls limit 10
17 03:59:32 UPDATE calls SET csat_score = NULL where csat_score = 0
18 03:59:41 SELECT * FROM calls limit 100
19 04:01:16 SELECT COUNT(*) AS rows_num FROM calls LIMIT 0, 1000
20 04:02:32 SELECT * FROM calls limit 100
```

Return rows - 32941

The screenshot shows the MySQL Workbench interface with a SQL query in the 'SQL Editor' pane. The query is 'SELECT COUNT(*) AS rows_num FROM calls;'. The 'Result Grid' at the bottom shows the output of the query, which is 32941.

```
40 • SELECT COUNT(*) AS rows_num FROM calls;
```

rows_num
32941

Return rows - 12

The screenshot shows the MySQL Workbench interface with a SQL query in the 'SQL Editor' pane. The query is 'SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';'. The 'Result Grid' at the bottom shows the output of the query, which is 12.

```
42 • SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';
```

cols_num
12

44 • `SELECT DISTINCT sentiment FROM calls;`

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	sentiment
▶	Neutral
	Very Positive
	Negative
	Very Negative
	Positive

45

46 • `SELECT DISTINCT reason FROM calls;`

47

48 • `SELECT DISTINCT channel FROM calls;`

49

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	reason
▶	Billing Question
	Service Outage
	Payments

47

48 • `SELECT DISTINCT channel FROM calls;`

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	channel
▶	Call-Center
	Chatbot
	Email
	Web

50

50 • `SELECT DISTINCT response_time FROM calls;`

51

52 • `SELECT DISTINCT call_center FROM calls;`

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	response_time
▶	Within SLA
	Above SLA
	Below SLA

51

52 • `SELECT DISTINCT call_center FROM calls;`

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	call_center
▶	Los Angeles/CA
	Baltimore/MD
	Denver/CO
	Chicago/IL

53

54 • `SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct`
55 `FROM calls GROUP BY 1 ORDER BY 3 DESC;`

56

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	sentiment	count(*)	pct
▶	Negative	11063	33.6
	Neutral	8754	26.6
	Very Negative	6026	18.3
	Positive	3928	11.9
	Very Positive	3170	9.6

56

```
57 • SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
58 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

59

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	reason	count(*)	pct
►	Billing Question	23462	71.2
	Payments	4749	14.4
	Service Outage	4730	14.4

59

```
60 • SELECT channel, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
61 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

62

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	channel	count(*)	pct
►	Call-Center	10639	32.3
	Chatbot	8256	25.1
	Email	7470	22.7
	Web	6576	20.0

64

```
63 • SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
64 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

65

```
66 • SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
67 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

	response_time	count(*)	pct
▶	Within SLA	20625	62.6
	Below SLA	8148	24.7
	Above SLA	4168	12.7

65

```
66 • SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
67 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

	call_center	count(*)	pct
▶	Los Angeles/CA	13734	41.7
	Baltimore/MD	11012	33.4
	Chicago/IL	5419	16.5
	Denver/CO	2776	8.4

68

```
69 • SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

	state	COUNT(*)
▶	California	3631
	Texas	3572
	Florida	2834
	New York	1786
	Virginia	1164
	Ohio	1160
	District of Columbia	1110
	Pennsylvania	1017
	Georgia	926
	Illinois	848
	North Carolina	765
	Colorado	742
	Alabama	738
	Arizona	737
	Indiana	736
	Minnesota	712
	Missouri	682
	Tennessee	664

Result 21 x

```
98 • SELECT DAYNAME(call_timestamp) as Day_of_call, COUNT(*) num_of_calls FROM calls group by 1 ORDER by 2 desc;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Day_of_call	num_of_calls			
Friday	5570			
Thursday	5481			
Wednesday	4449			
Tuesday	4408			
Saturday	4403			
Monday	4334			
Sunday	4296			

```
73  
74 • SELECT MIN(csat_score) AS min_score, MAX(csat_score) AS max_score, ROUND(AVG(csat_score),1) AS avg_score  
75 FROM calls WHERE csat_score != 0; # MySql added 0 to blank rows. But the min is 1.
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
min_score	max_score	avg_score		
1	10	5.5		

```
/b  
77 • SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
earliest_date	most_recent			
2020-10-01	2020-10-31			

```
79 • SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_duration FROM calls;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
min_call_duration	max_call_duration	avg_call_duration		
5	45	25.0212		

80

```
81 • SELECT call_center, response_time, COUNT(*) AS count
82 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
```

83

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	call_center	response_time	count
▶	Baltimore/MD	Within SLA	6855
	Baltimore/MD	Below SLA	2768
	Baltimore/MD	Above SLA	1389
	Chicago/IL	Within SLA	3361
	Chicago/IL	Below SLA	1361
	Chicago/IL	Above SLA	697
	Denver/CO	Within SLA	1741
	Denver/CO	Below SLA	692
	Denver/CO	Above SLA	343
	Los Angeles/CA	Within SLA	8668
	Los Angeles/CA	Below SLA	3327
	Los Angeles/CA	Above SLA	1739

83

```
84 • SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	call_center	AVG(call_duration_minutes)
▶	Chicago/IL	25.0626
	Los Angeles/CA	25.0532
	Denver/CO	25.0166
	Baltimore/MD	24.9620

85

```
86 • SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	channel	AVG(call_duration_minutes)
▶	Email	25.0988
	Call-Center	25.0462
	Web	25.0224
	Chatbot	24.9178

87

88 • `SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;`

89

90 • `SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;`

91

92 • `SELECT state, sentiment , COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;`

93

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	state	COUNT(*)
►	California	3631
	Texas	3572
	Florida	2834
	New York	1786
	Virginia	1164
	Ohio	1160
	District of Columbia	1110
	Pennsylvania	1017
	Georgia	926
	Illinois	848
	North Carolina	765
	Colorado	742
	Alabama	738
	Arizona	737
	Indiana	736
	Minnesota	712
	Missouri	682
	Tennessee	664
	Washington	663
	Louisiana	627
	Michigan	612

Result 28 ×

89

90 • `SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;`

91

92 • `SELECT state, sentiment , COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;`

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	state	reason	COUNT(*)
▶	Alabama	Billing Question	538
	Alabama	Payments	105
	Alabama	Service Outage	95
	Alaska	Billing Question	93
	Alaska	Payments	25
	Alaska	Service Outage	28
	Arizona	Billing Question	507
	Arizona	Payments	119
	Arizona	Service Outage	111
	Arkansas	Billing Question	152
	Arkansas	Payments	29
	Arkansas	Service Outage	23
	California	Billing Question	2588
	California	Payments	501
	California	Service Outage	542
	Colorado	Billing Question	516
	Colorado	Payments	132
	Colorado	Service Outage	94
	Connect...	Billing Question	275
	Connect...	Payments	69
	Connect...	Service Outage	64

Result 29 ×

91

92 • `SELECT state, sentiment , COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;`Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	state	sentiment	COUNT(*)
▶	Alabama	Negative	265
	Alabama	Neutral	200
	Alabama	Very Negative	129
	Alabama	Positive	74
	Alabama	Very Positive	70
	Alaska	Negative	45
	Alaska	Neutral	32
	Alaska	Very Negative	28
	Alaska	Positive	25
	Alaska	Very Positive	16
	Arizona	Negative	235
	Arizona	Neutral	204
	Arizona	Very Negative	131
	Arizona	Positive	94
	Arizona	Very Positive	73
	Arkansas	Negative	73
	Arkansas	Neutral	50
	Arkansas	Very Negative	39
	Arkansas	Very Positive	22
	Arkansas	Positive	20
	California	Negative	1255

Result 30 ×

94

```
SELECT state, AVG(csatscore) as avg_csatscore FROM calls WHERE csatscore != 0 GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [iA](#)

state	avg_csatscore
Vermont	6.5000
North Dakota	6.3714
Wyoming	6.0000
Massachusetts	5.9070
Rhode Island	5.8889
Hawaii	5.8727
Mississippi	5.8286
Idaho	5.8182
Washington	5.7574
Louisiana	5.7448
Illinois	5.7138
Georgia	5.6878
Utah	5.6733
Pennsylvania	5.6684
West Virginia	5.6321
Ohio	5.6282
Florida	5.6217
Tennessee	5.6044
Oklahoma	5.6044
Texas	5.5877
Kansas	5.5749

Result 31 x

95

```
96 • SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [iA](#)

sentiment	AVG(call_duration_minutes)
Negative	25.2618
Neutral	24.9398
Very Negative	24.9391
Positive	24.8620
Very Positive	24.7593

99

100 • `SELECT call_timestamp, max(call_duration_minutes) OVER(partition by call_timestamp) AS max_call_duration FROM calls GROUP by 1 ORDER BY 2 DESC;`

Result Grid   Filter Rows:

Export: 

Wrap Cell Content: 

	call_timestamp	max_call_duration
▶	2020-10-04	45
	2020-10-22	45
	2020-10-26	41
	2020-10-13	39
	2020-10-30	37
	2020-10-21	37
	2020-10-03	37
	2020-10-14	37
	2020-10-12	35
	2020-10-09	35
	2020-10-25	34
	2020-10-16	31
	2020-10-20	30
	2020-10-19	30
	2020-10-02	30
	2020-10-18	28
	2020-10-08	27
	2020-10-27	26
	2020-10-28	25
	2020-10-05	23
	2020-10-24	22
	2020-10-06	22
	2020-10-10	20
	2020-10-01	19
	2020-10-15	19
	2020-10-11	18
	2020-10-29	17
	2020-10-23	14
	2020-10-31	13
	2020-10-17	12
	2020-10-07	12